

AI in Robotics

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Humanoid Robots



Apollo



Agility Robotics



Figure 01



UniTree



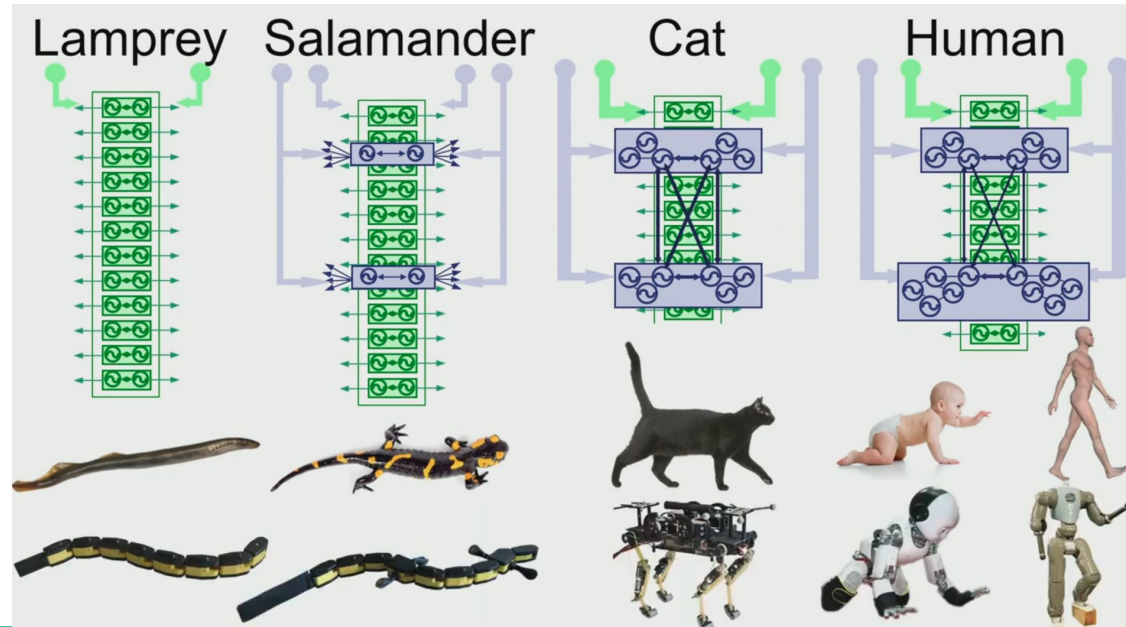
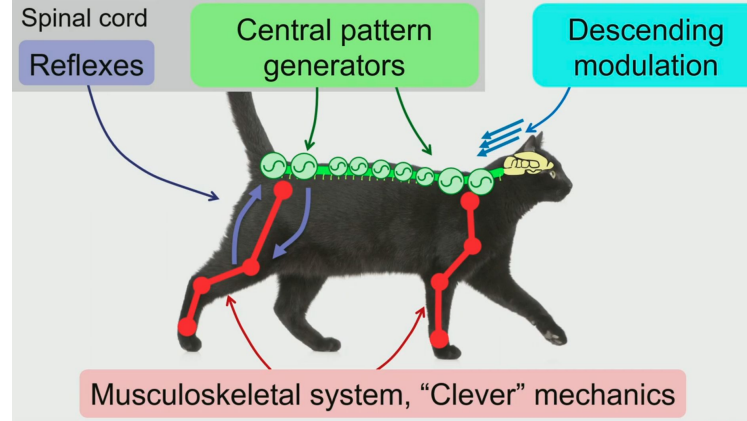
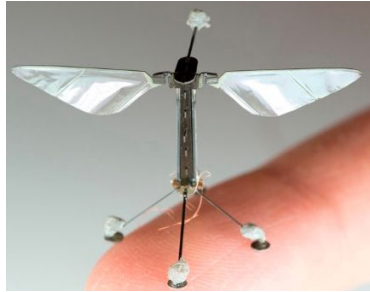
Phoenix



Testa Optimus



Animal-inspired robots



RoboTaxi - Self-Driving Vehicles



Robot Taxi
@Sanfrancisco

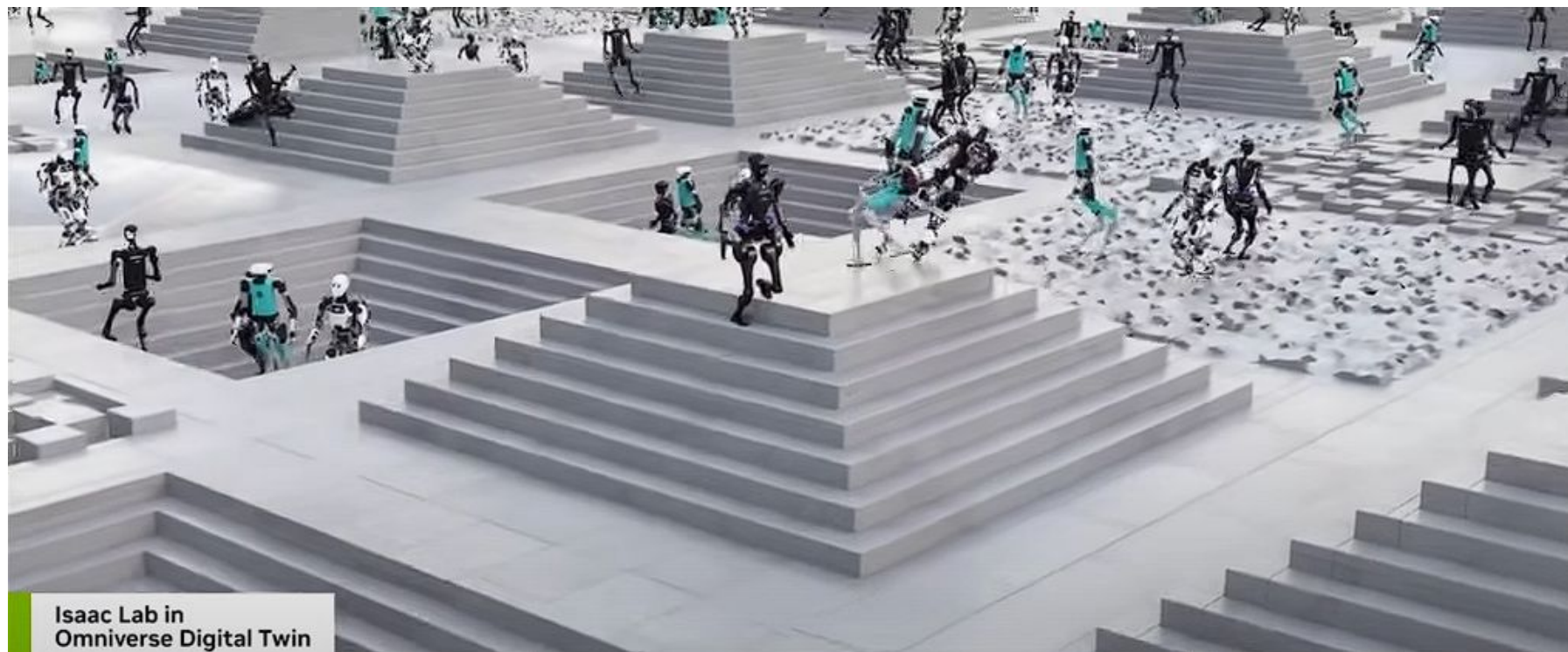


AutoX - Alibaba
China

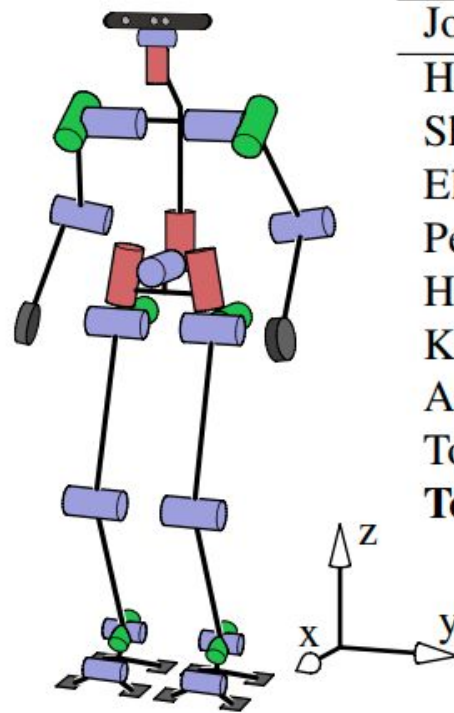


Hyundai SDV
South Korea

3D Simulation: IsaacSim (Nvidia)



Robot Kinematic

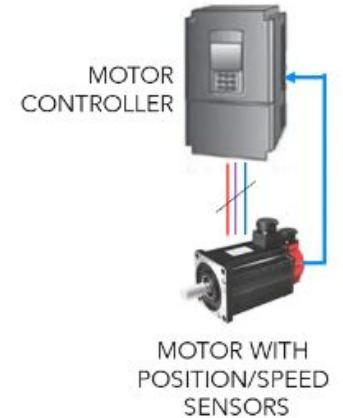
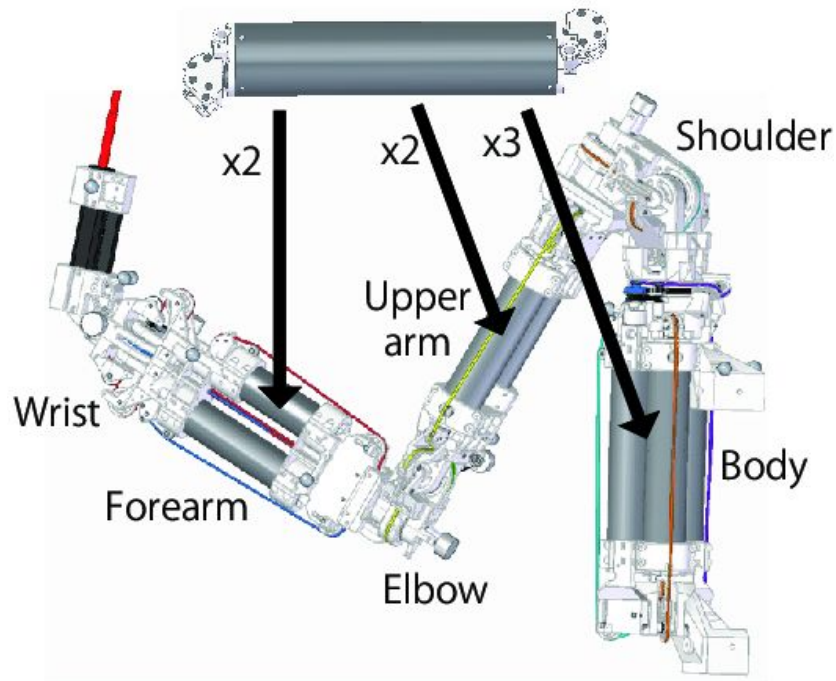


Joint	DoF
Head	2
Shoulder	2
Elbow	1
Pelvis	2
Hip	3
Knee	1
Ankle	2
Toe	1
Total	24

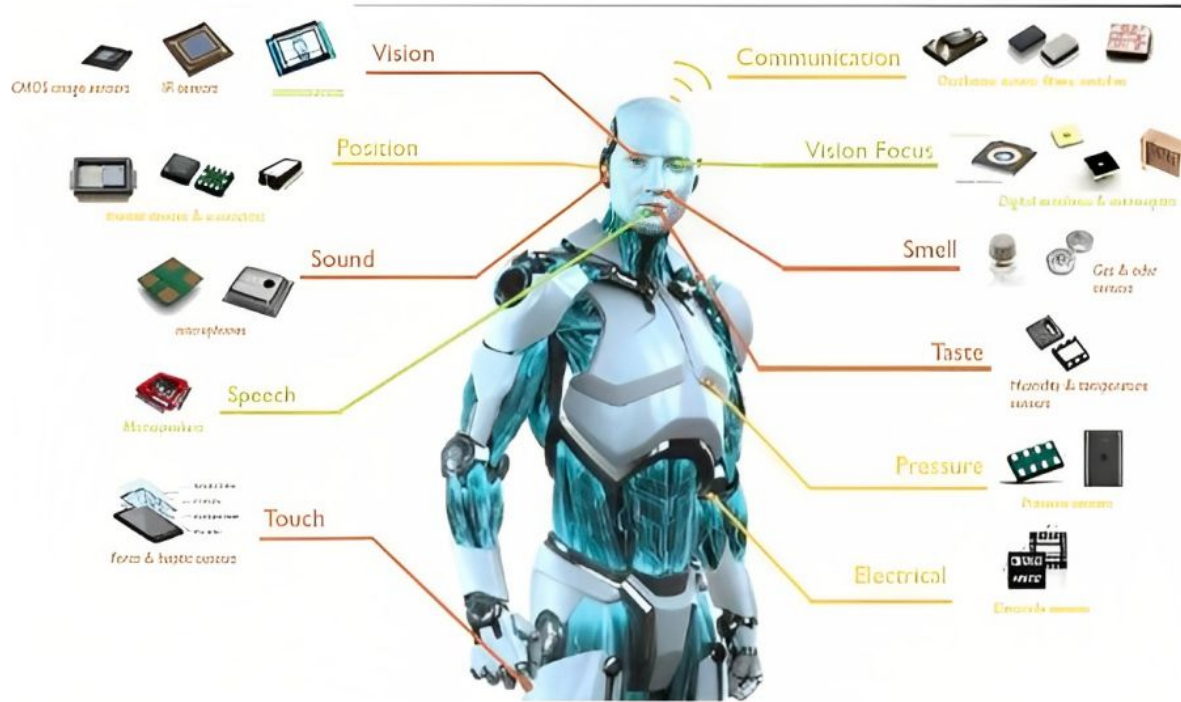
Kinematic optimization for bipedal robots: A framework for real-time collision avoidance
Hukderabdt et al, 2017

Robot Actuators (motor, hydraulic,..)

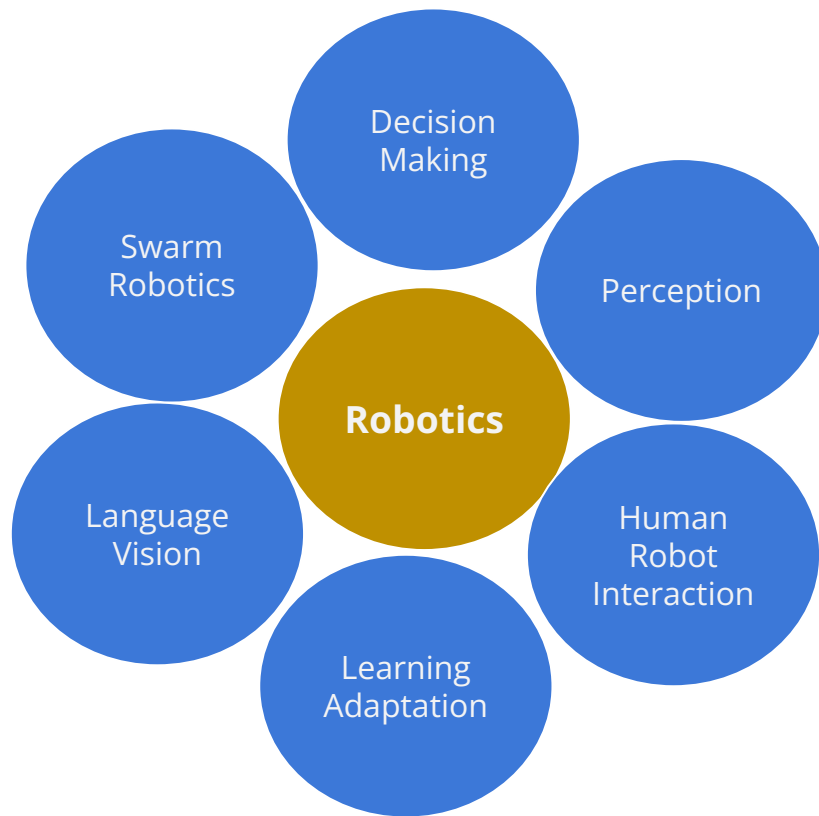
Developed Hybrid Actuators



Robot Sensors - Multimodalities



Robot Learning



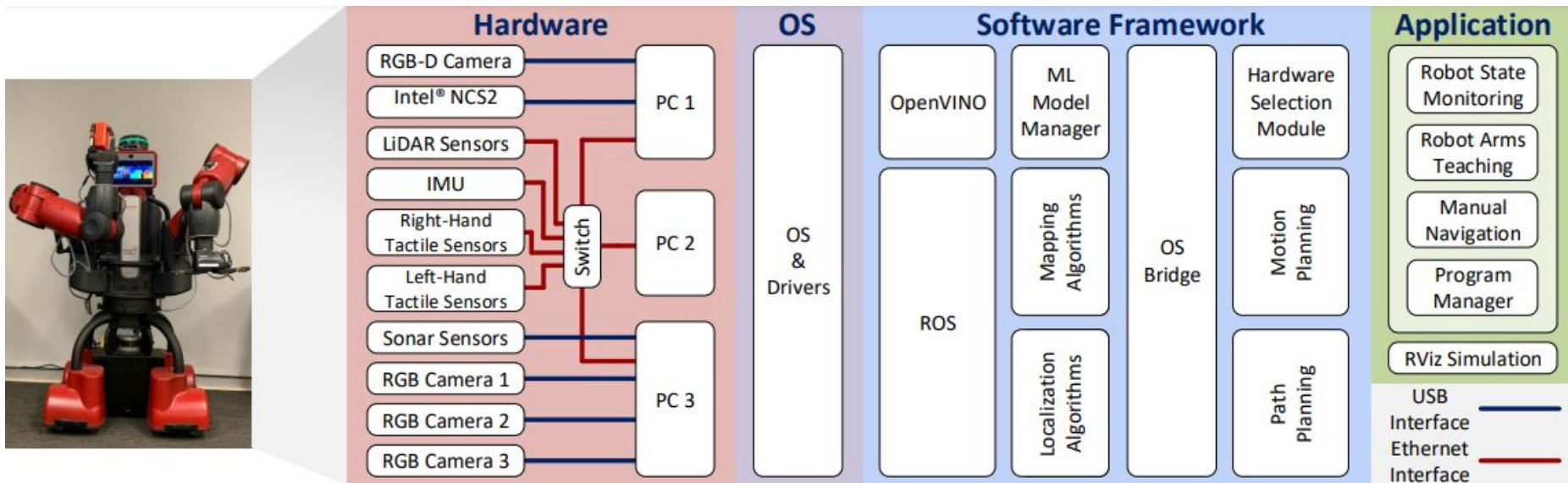
Autonomy and Decision Making

- Motion Planning
- Task Planning
- Reactive Decision-making to immediate stimuli
- Deliberative Decision Making:
 - Probabilistic reasoning
 - Markov Decision Process (MDP)
 - Partially Observable MDP

Perceptions

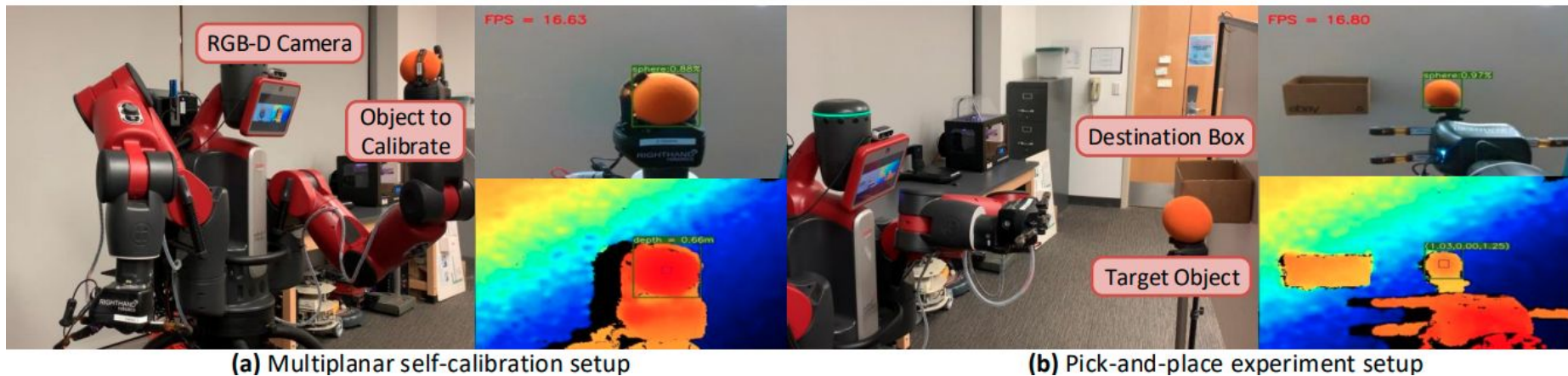
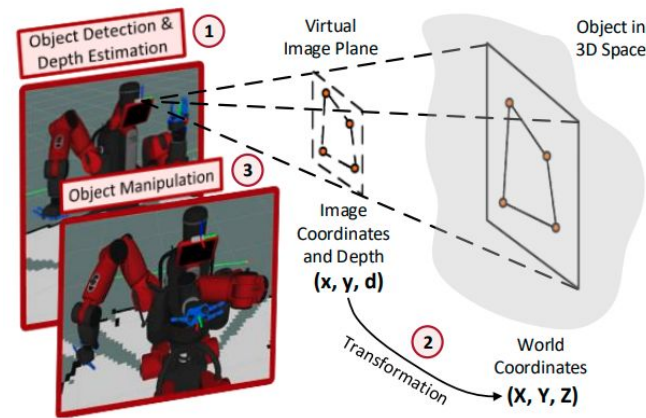
- Understanding sensing data
- 2D & 3D Perception
- Vision
- LiDAR
- Sounds
- Fusions
- Etc, ..

Case study: Hardware Software Framework



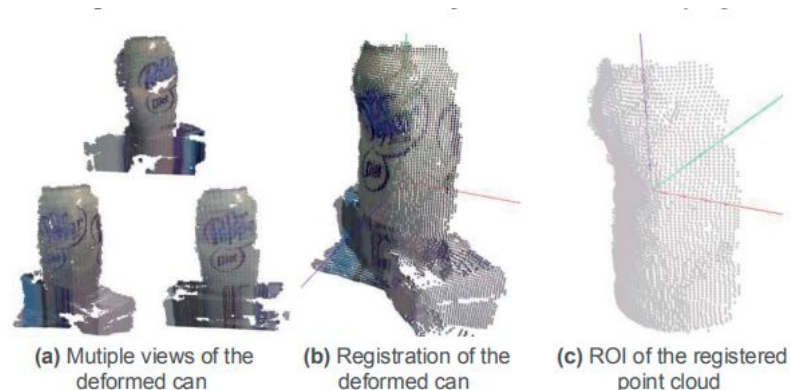
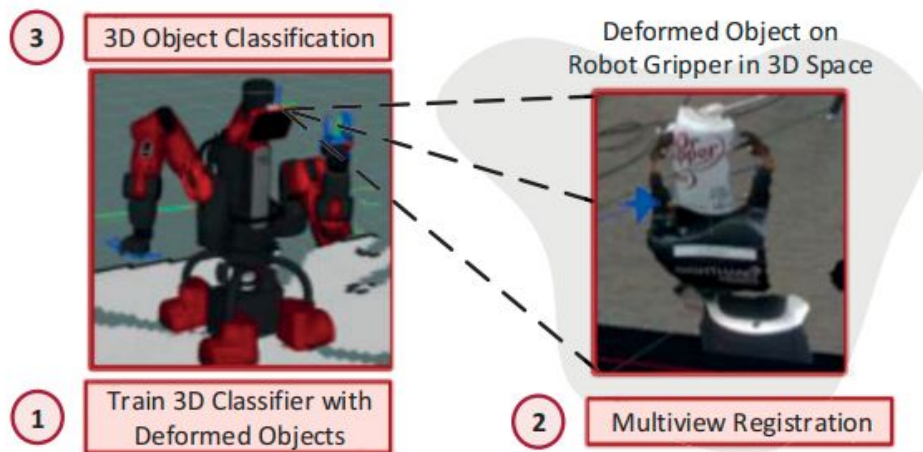
PerFc : An Efficient 2D and 3D Perception Software-Hardware Framework for Mobile Cobot
Tuan Dang, Khang Nguyen, Manfred Huber, FLAIRS 2023

Vision-based Self-Calibration

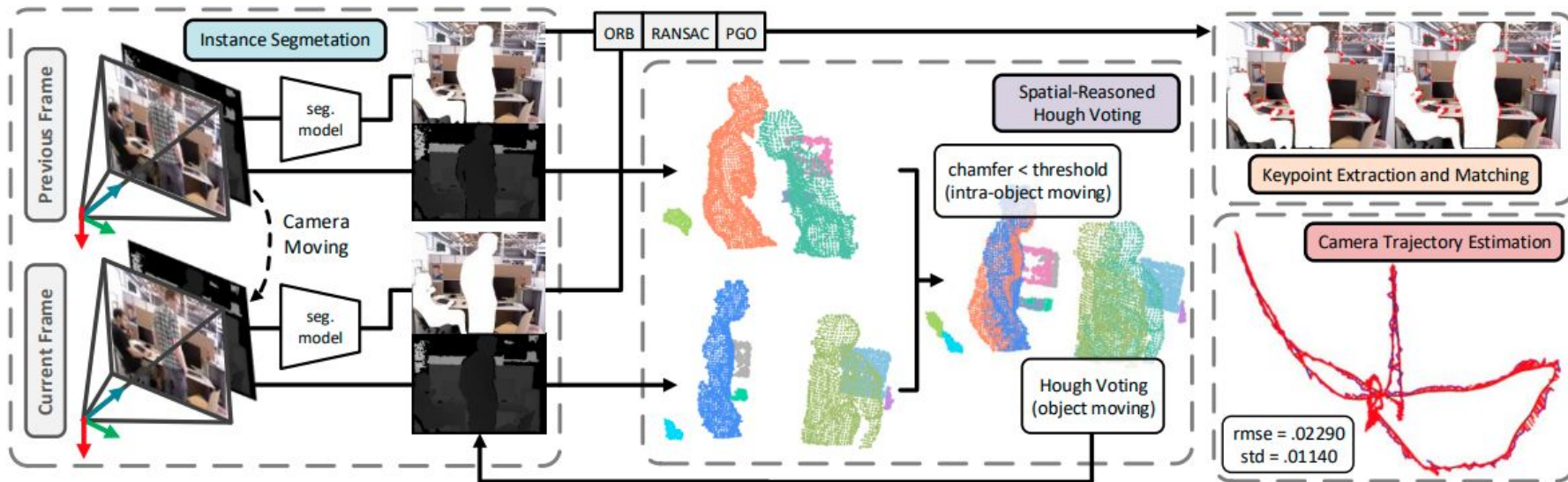


Multiplanar Self-Calibration for Mobile Cobot 3D Object Manipulation using 2D Detectors and Depth Estimation
Tuan Dang, Khang Nguyen, Manfred Huber, IROS 2023

Multi-view 3D Object Recognition



Simultaneous Localization And Mapping (SLAM)

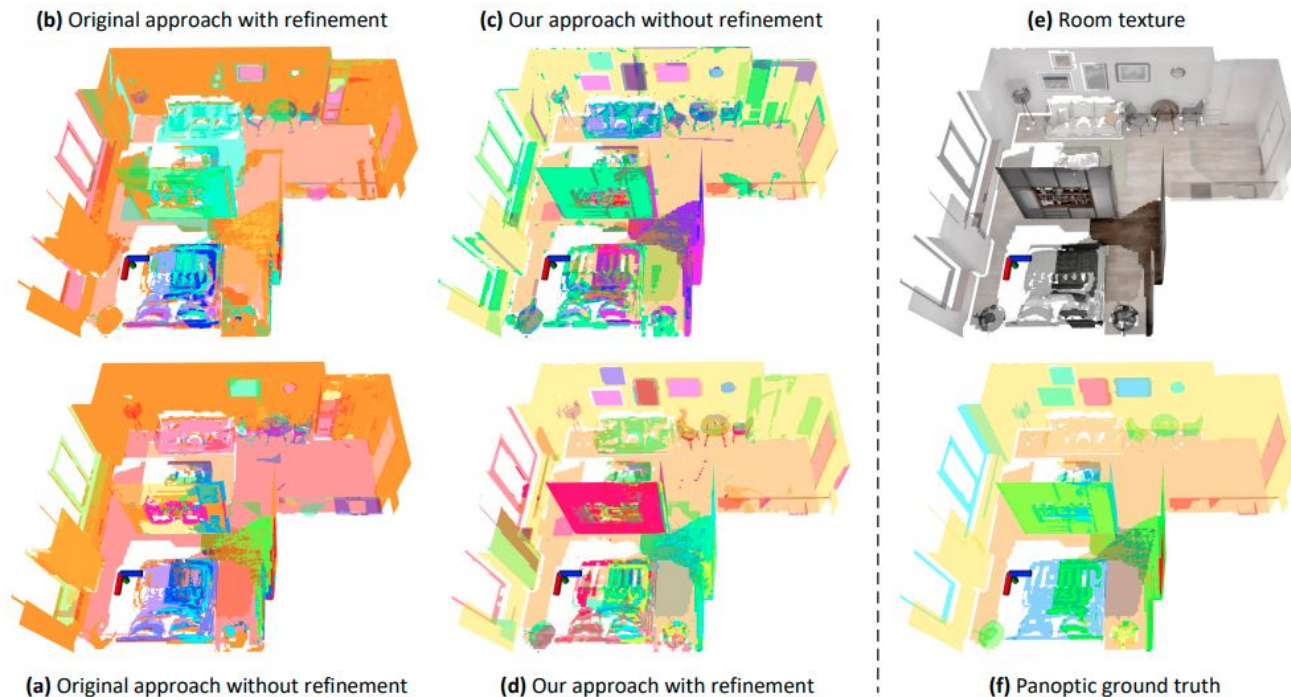


V3D-SLAM : Robust RGB-D SLAM in Dynamic Environments with 3D Semantic Geometry Voting
Tuan Dang, Khang Nguyen, Manfred Huber, IROS 2024

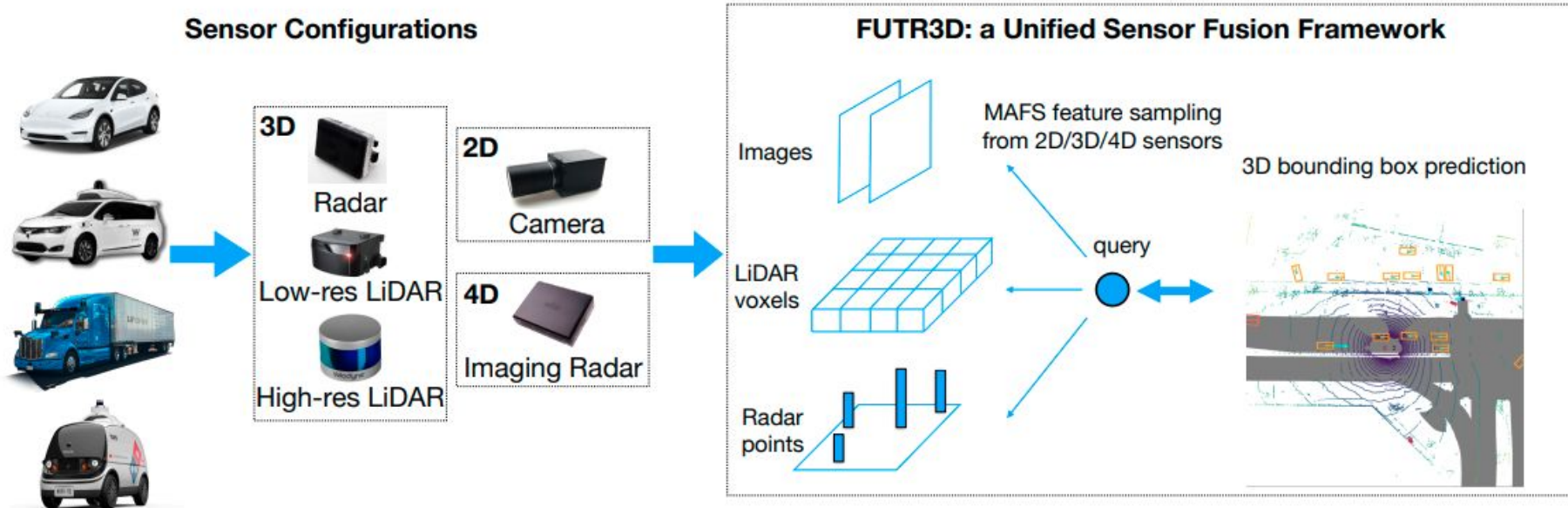
Motion Trajectory Estimation



3D Reconstruction (Structure from Motion - SfM)

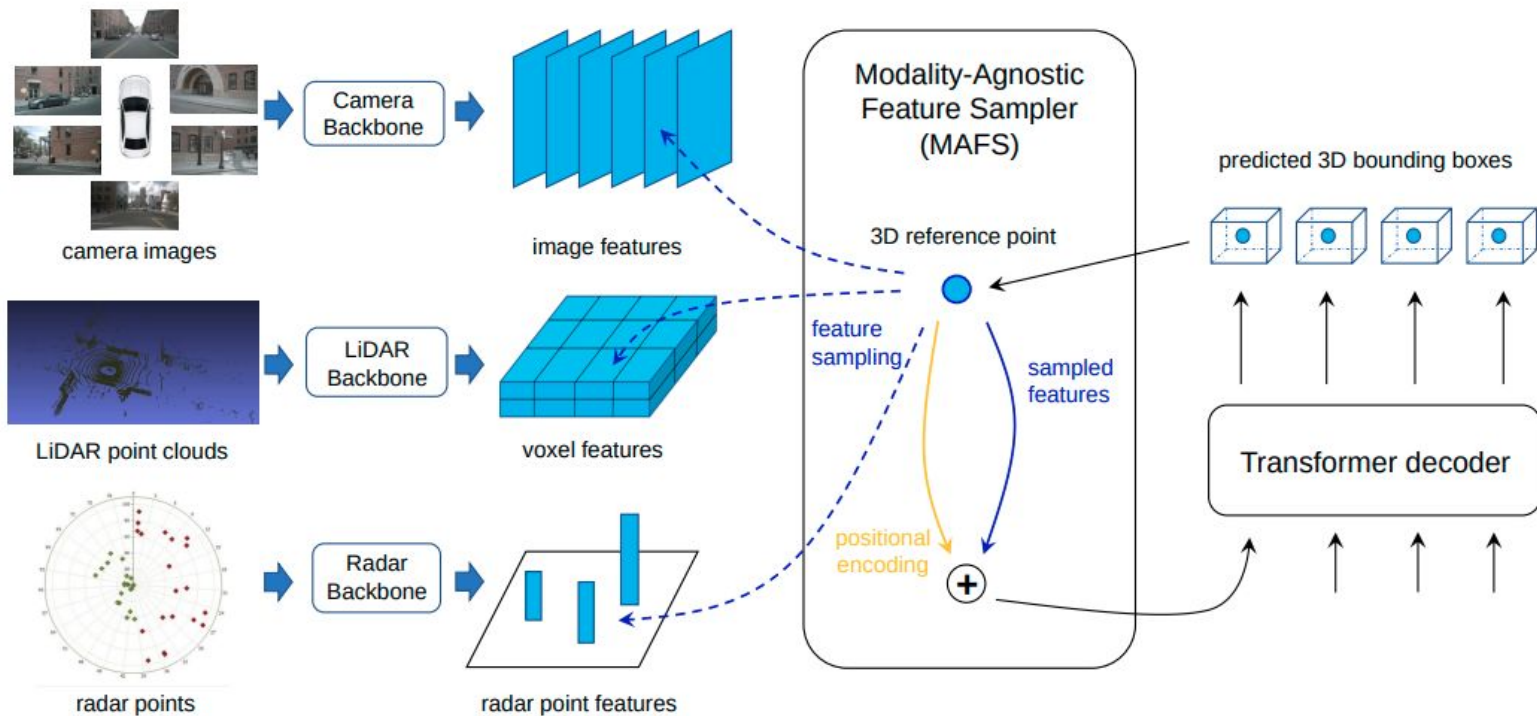


Object Detections using Multi-modalities



FUTR3D: A Unified Sensor Fusion Framework for 3D Detection
Chen et al, 2023

Network Architectures



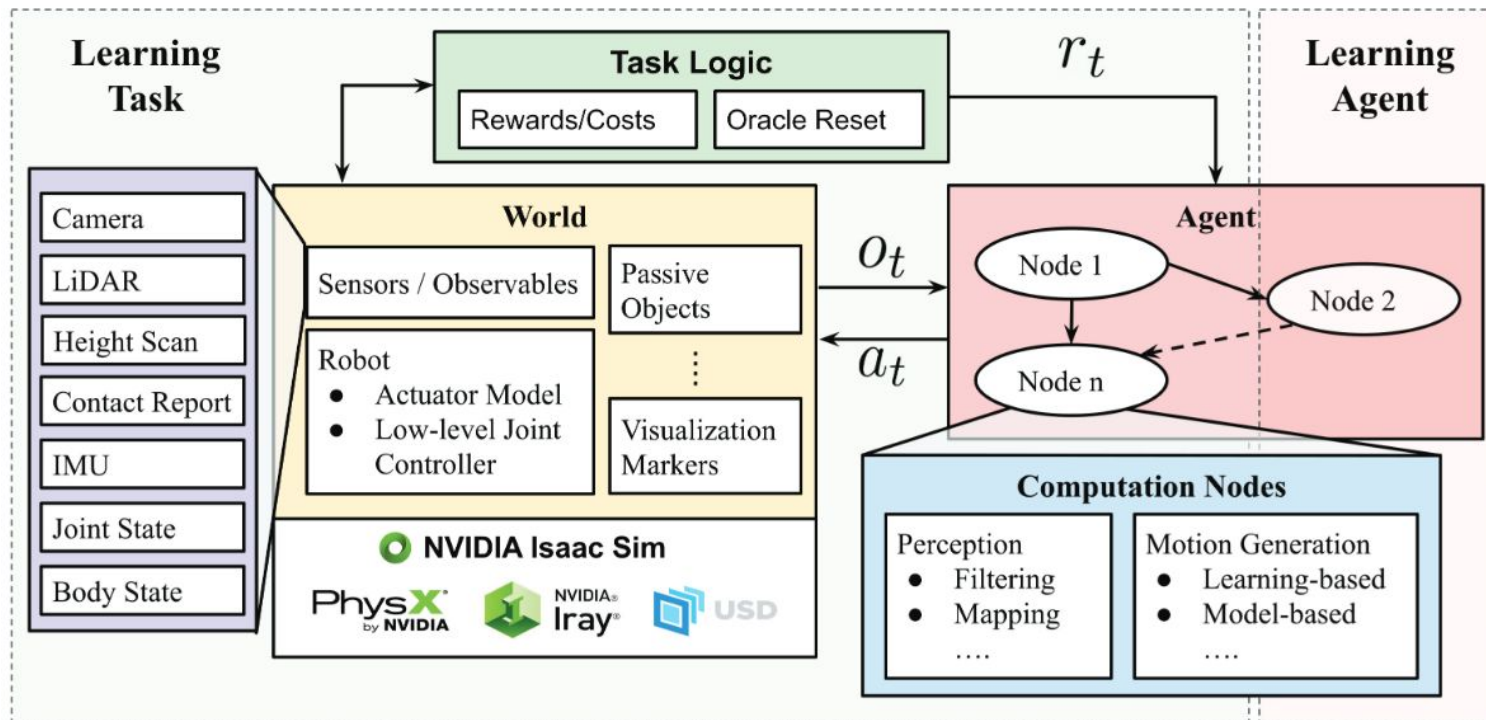
Human Robot Interaction

- Natural Language Processing (NLP)
- Emotion and Sentiment Recognition
- Gesture Recognition
- Learn from Interaction
- Applications:
 - Healthcare Assistive Robots
 - Service Robots
 - Educational and Tutoring Robots

Adaptive Learning

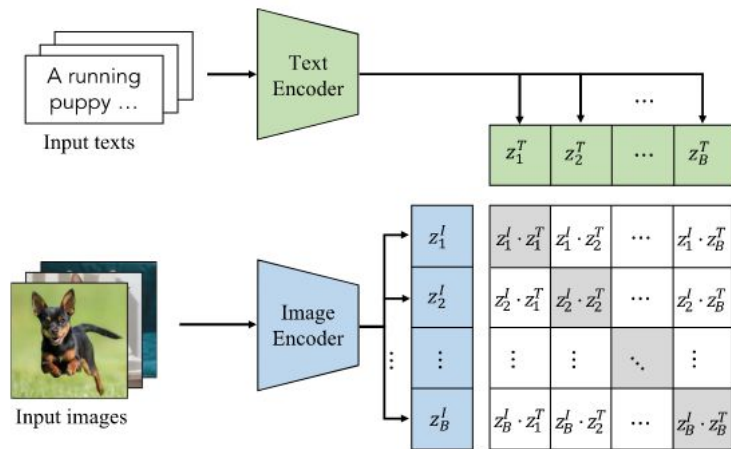
- Adapt to the changes in the environments
- What is adaptive learning:
 - Learn from **unlabeled data** collecting during interactions with the environments
 - Feedback-based Learning: use signals/stimuli to guide robots to learn a new task.
- Type of learnings
 - **Reinforcement Learning:** use Reward is the feedback signals to improve the decision making by self-explore the action space to maximize the long-term reward.
 - **Online Learning:** update models constantly
 - **Transfer Learning:** transfer knowledge between similar tasks or use distillation network
 - **Imitation Learning:** replicate the demonstrated tasks
 - Multi-task Learning: learn multiple tasks simultaneously

Learn in Simulation

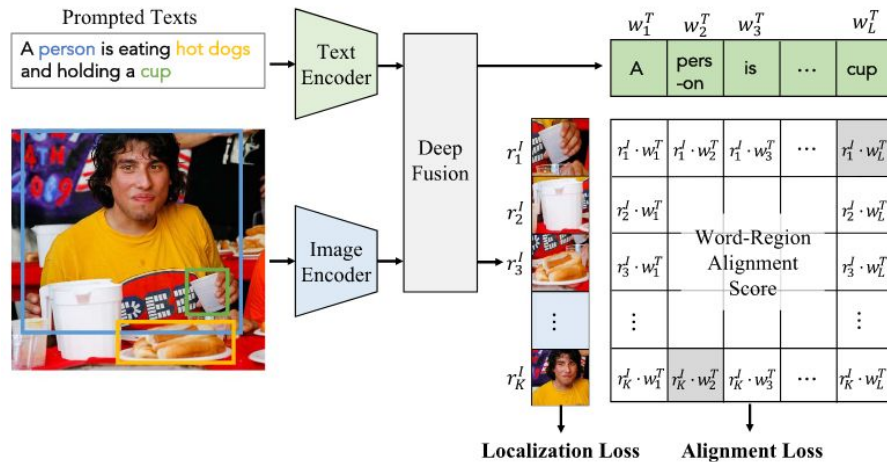


Vision Language Models (VLMs)

- Learn the relationships between visual content and corresponding textual description.
- Tasks: Image Captioning, Visual Question Answering (VQA)



CLIP

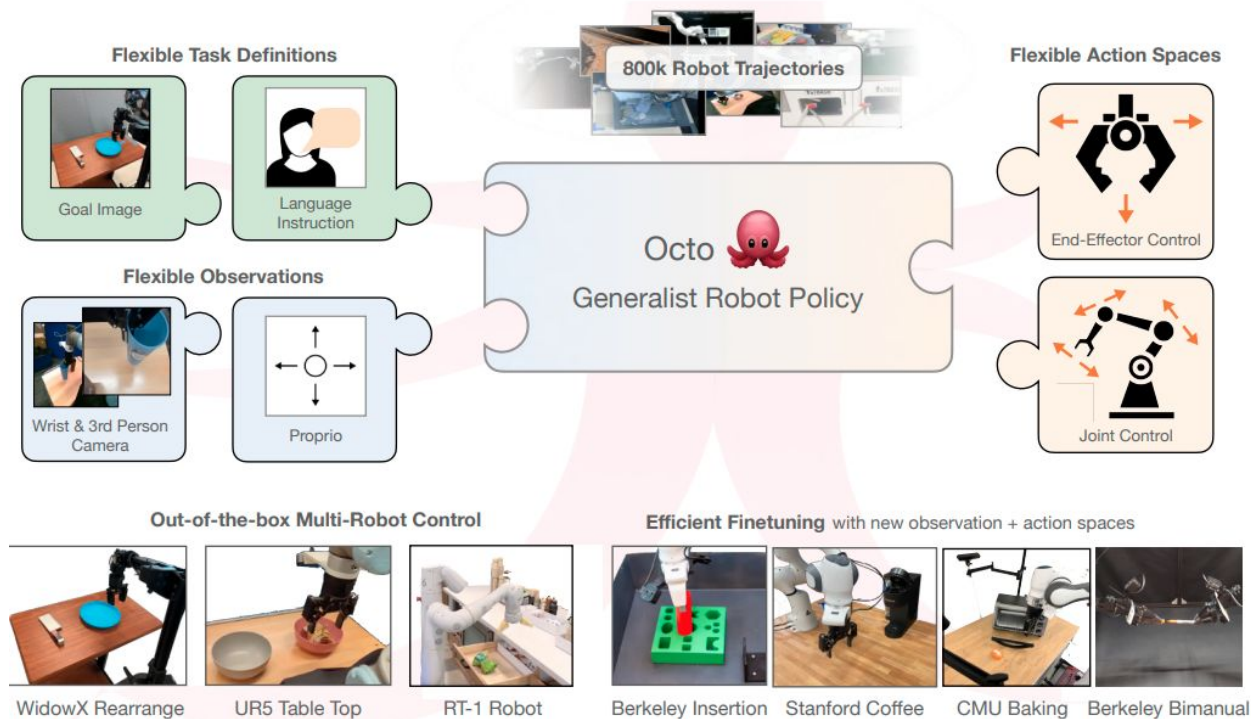


GLIP

Case Study: Octo

```
from octo.model import OctoModel
```

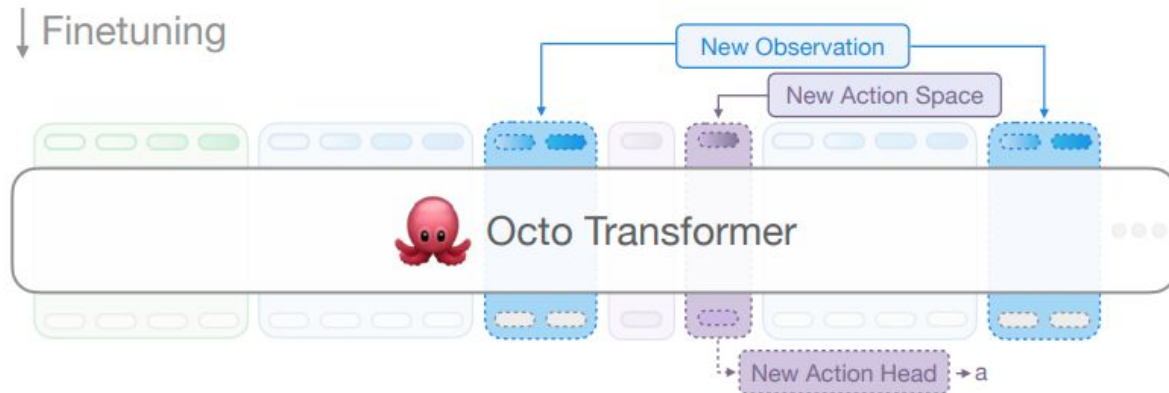
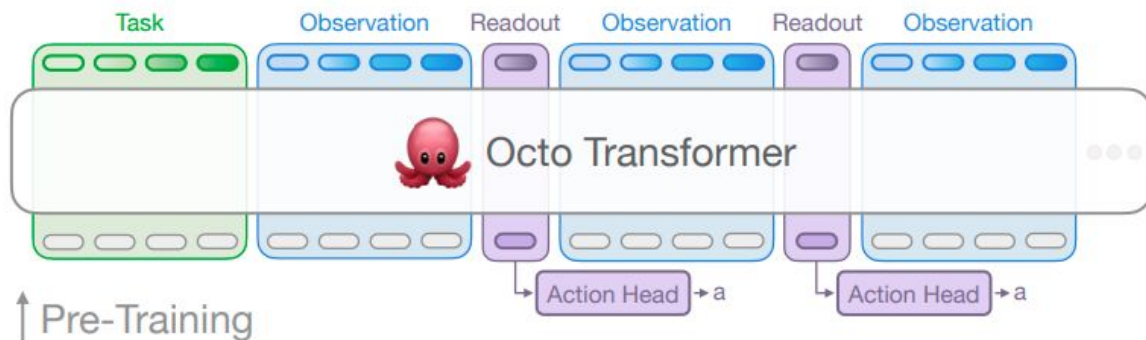
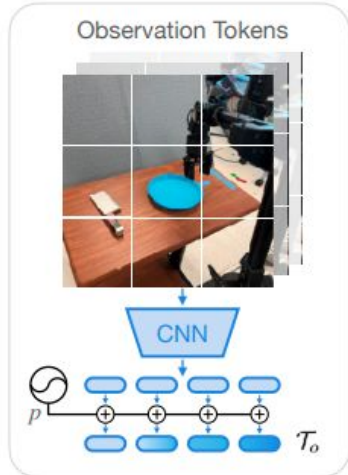
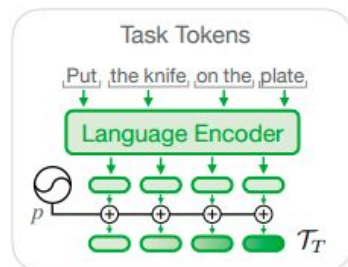
```
model = OctoModel.load_pretrained("hf://rail-berkeley/octo-small-1.5")  
task = model.create_tasks(texts=["pick up the spoon"])  
action = model.sample_actions(observation, task, rng=jax.random.PRNGKey(0))
```



Octo: An Open-Source Generalist Robot Policy, <https://arxiv.org/pdf/2405.12213>

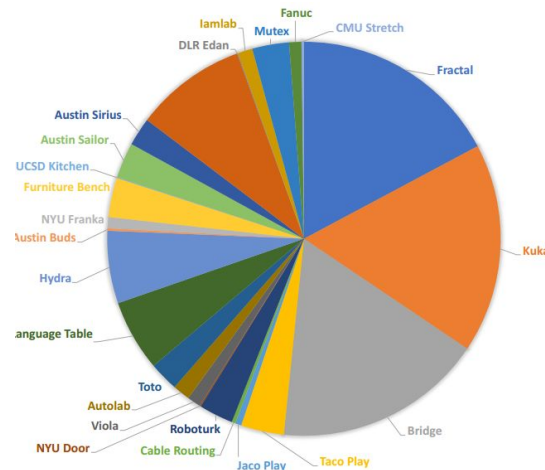
<https://github.com/octo-models/octo>

Model Architecture



Training

- Datasets:
 - 25 datasets from Open X-Embodiment
 - Heterogeneous
- Conditional Diffusion Decoding
 - **Conditioning Input:** text-to-images generation, class-conditioned generation,...
 - **Guided Denoising Process:** use conditional information guides neural network each step



$$x^{k-1} = \alpha(x^k - \gamma \epsilon_{\theta}(x^k, e, k) + \mathcal{N}(0, \sigma^2 I))$$

Denoising
Network

Gaussian
Noise

Hyper-parameters

Swarm Robotics

- Coordination of multiple robots
- Decentralized Control
- **Distributed RL**
- Traditional algorithms:
 - Genetic Algorithm (GA)
 - Finite State Machine (FSM)
 - Consensus Algorithm (CA)
 - Particle Swarm Optimization (PSO)



AI Challenges in Robotics

- Trust: Accuracy, Reliability,
- Ethical Decision-Making
- Bias and Fairness

Open Question

If robots can do most of human work,
what can we do?

Learn More

- Why Don't We Have Better Robots Yet? | Ken Goldberg | TED
 - <https://www.youtube.com/watch?v=cUdl-Cp-LWw>
-

Join us

- Reinforcement Learning: Learn from Simulation
- Email: tuan.dang@uta.edu or dangthanhtuanit@gmail.com
- Support: Nvidia RXT 4090 and A6000

